

FRIT for Systems with Dead-Zone and Its Application to Ultrasonic Motors

Yuji Wakasa* Member, Shinji Kanagawa* Non-member
 Kanya Tanaka* Senior Member, Yuki Nishimura* Member

(Manuscript received Aug. 23, 2010, revised Dec. 24, 2010)

Ultrasonic motors (USMs) intrinsically have a dead-zone property which is sensitive to load changes. This paper proposes a fictitious reference iterative tuning (FRIT) method for systems with a dead-zone property such as USMs. The standard FRIT method is basically developed for linear systems and may not give a satisfactory control performance for noninvertible nonlinear systems including USMs. In contrast, the proposed FRIT method can achieve such a performance by introducing a right inverse of a dead-zone function as a dead-zone compensator. In the optimization process of FRIT, the so-called covariance matrix adaptation evolution strategy (CMA-ES) is used for simultaneously searching a dead-zone parameter as well as controller parameters. CMA-ES is a kind of stochastic multi-point search techniques and is effective for nondifferentiable and nonconvex optimization problems. Experimental results for a USM are given to show the effectiveness of the proposed method.

Keywords: Fictitious reference iterative turning, PID control, Ultrasonic motor, Dead-zone, Covariance matrix adaptation evolution strategy

1. Introduction

The feature of fictitious reference iterative tuning (FRIT) is to tune controller parameters directly based on input/output data obtained from an only one-shot closed-loop experiment⁽¹⁾. Since FRIT does not need any complicated modeling processes, it has received much attention recently as a practical control system design method⁽²⁾.

On the other hand, ultrasonic motors (USMs) have excellent electromagnetic compatibility because, in principle, they generate no electromagnetic noise, and are not influenced by electromagnetic fields⁽³⁾. Although, for this reason, USMs are promising devices in fields such as medical and welfare, they cannot be easily modeled from physical analysis due to its friction force. Also, USMs have a sensitive dead-zone to temperature and load changes, which leads to deterioration of control performance. Therefore, a simple controller tuning method such as FRIT is desirable. In general, however, the optimality of FRIT does not hold for systems with a dead-zone, and therefore, FRIT may not be effective for USMs.

To resolve the above problem, in this paper, we propose an FRIT method combined with dead-zone compensation. In the optimization process of the FRIT method, we use the so-called covariance matrix adaptation evolution strategy (CMA-ES)⁽⁴⁾ for simultaneously

searching a dead-zone parameter as well as controller parameters. CMA-ES is a kind of stochastic multi-point search techniques and is effective for nondifferentiable and nonconvex optimization problems.

The rest of this paper is organized as follows. In Section 2, we first present a framework of feedback control and dead-zone compensation. In Section 3, we propose an FRIT method for tuning controller parameters and a dead-zone parameter. Then, in Section 4, we describe the CMA-ES algorithm for solving the optimization problem in the FRIT procedure. Section 5 presents the experimental USM control system and its dead-zone property. Section 6 shows experimental results, and finally, Section 7 provides our conclusions and scope for future research.

2. Problem Formulation

Consider a single-input and single-output system consisting of the series connection of a dead-zone D_a and a linear system $G(z)$ as shown in Fig. 1. We assume that the dead-zone D_a has a dead-zone interval $[-a, a]$ ($a > 0$) and its input/output relation can be expressed as

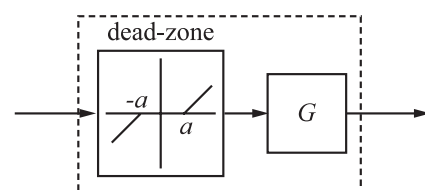


Fig. 1. A system with a dead-zone that captures the behavior of USMs.

* Graduate School of Science and Engineering, Yamaguchi University
 2-16-1, Tokiwadai, Ube, Yamaguchi 755-8611, Japan

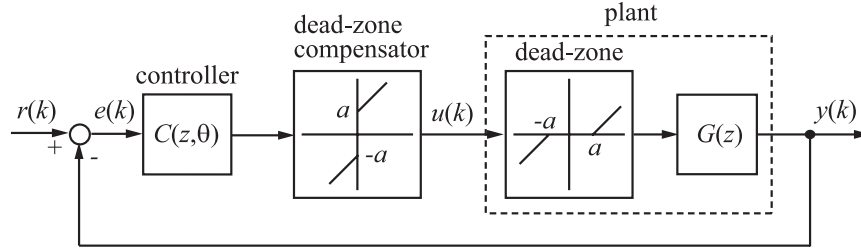


Fig. 2. Configuration of the control system.

$$D_a(u) = \begin{cases} u + a & u < -a \\ 0 & -a \leq u \leq a \\ u - a & u > a \end{cases} \quad (1)$$

The above system configuration is motivated by characteristics of USMs and is shown to be useful to control USMs adequately in ⁽⁵⁾.

For the dead-zone D_a , we define the following function.

$$\hat{D}_a(\hat{u}) = \begin{cases} \hat{u} - a & \hat{u} < 0 \\ 0 & \hat{u} = 0 \\ \hat{u} + a & \hat{u} > 0 \end{cases} \quad (2)$$

The function $\hat{D}_a$ is a right inverse function of D_a since $\hat{u} = D_a(\hat{D}_a(\hat{u}))$ holds for any $\hat{u} \in \mathbb{R}$.

In this paper, we compensate the dead-zone by using the right inverse function $\hat{D}_a$ as a dead-zone compensator and consider a configuration of a feedback control system as shown in Fig. 2. In the figure, C denotes a controller such as a proportional-integral-derivative (PID) controller and θ denotes a parameter vector to be tuned in the controller. Also, $u(k)$, $y(k)$, $r(k)$, and $e(k)$ denote the control input, control output, reference signal, and tracking error, respectively. For instance, the transfer function of the PID controller can be expressed as

$$C(z, \theta) = \frac{K_P(1 - z^{-1}) + K_I + K_D(1 - z^{-1})^2}{1 - z^{-1}}, \dots \quad (3)$$

where K_P , K_I , and K_D are the proportional, integral, and derivative gains, respectively, and $\theta = (K_P, K_I, K_D)^T$ includes the PID gains to be tuned.

Note that we can cancel the dead-zone property with the dead-zone compensator $\hat{D}_a$ completely if an actual dead-zone property can be expressed as D_a . In such an ideal case, therefore, all we have to do is to control the linear system G appropriately by the controller C , which may lead to an easy and effective control method. Although, in practice, there should be some modeling errors for the plant, we can expect that the above configuration achieves a satisfactory control performance as in ⁽⁵⁾ and ⁽⁶⁾.

Also note that the output of the dead-zone compensator can be vibrated for a small output of the controller. In most cases, however, $G(z)$ in the plant is strictly proper, and therefore, the control output cannot be influenced so much by such vibration. However, this dead-zone compensator should not be used for a plant in which a vibrated control input should be avoided.

In ⁽⁵⁾, a dead-zone property has to be modeled by some offline experiments. Although a dead-zone property can be estimated online in ⁽⁶⁾, a periodical reference signal is used to tune a dead-zone parameter a as well as a controller parameter θ . Such a periodical reference signal can be regarded as a requirement of multiple experiments.

In contrast, FRIT can give a good performance through just a one-shot experiment, which is a good feature from the practical viewpoints. However, the standard FRIT method has been developed for linear systems theoretically ⁽¹⁾⁽²⁾, and it is not clear whether the standard FRIT method is still effective for systems with a dead-zone. Although it is expected that some characteristics of the FRIT method such as the optimality are also satisfied for nonlinear systems as in the case of VRFT in ⁽⁷⁾, the proof of the optimality requires a kind of invertibility of the plant which is not satisfied for the plant with the abovementioned dead-zone.

Motivated from the above considerations, we aim to develop a tuning method of a dead-zone parameter a as well as a controller parameter θ based on FRIT.

3. FRIT under Dead-Zone Compensation

In FRIT, we have to calculate the fictitious reference signal $\tilde{r}$ which is different between the proposed framework introducing the dead-zone compensation and the standard FRIT shown in ⁽¹⁾. Suppose that there is no dead-zone compensation (*i.e.*, $a = 0$) and that we obtain input/output data $u_0(k)$, $y_0(k)$, $k = 1, \dots, N$, for an initial controller parameter θ_0 and a reference signal $r(k)$. Then, in the standard FRIT, the fictitious reference signal is calculated by

$$\tilde{r}(\theta, k) = C(z, \theta)^{-1}u_0(k) + y_0(k). \quad (4)$$

In contrast, the fictitious reference signal in our framework is calculated as follows:

$$\tilde{r}(x, k) = C(z, \theta)^{-1}D_a(u_0(k)) + y_0(k), \quad (5)$$

where $x = (\theta^T, a)^T$. Here, note that D_a is included in the calculation of the fictitious reference signal.

We summarize the FRIT procedure under dead-zone compensation as follows.

FRIT under Dead-Zone Compensation

Step 1. Set initial PID gains θ_0 , a reference signal $r(k)$, $k = 1, \dots, N$, and a reference model $M(z)$.

Step 2. From a closed-loop experiment that is performed without dead-zone compensation, we obtain

the input/output data $u_0(k)$, $y_0(k)$, $k = 1, \dots, N$, for the reference signal $r(k)$.

Step 3. According to (5), calculate the fictitious reference signal $\tilde{r}(\mathbf{x}, k)$. Then, for the tuning parameter $\mathbf{x} = (\boldsymbol{\theta}^T, a)^T$, find the optimal (or suboptimal) parameter $\mathbf{x}$ to minimize the performance index

$$J(\mathbf{x}) = \sum_{k=1}^N (y_0(k) - M(z)\tilde{r}(\mathbf{x}, k))^2 \dots \dots \dots (6)$$

to obtain the optimal (or suboptimal) PID gains $\boldsymbol{\theta}^*$ and dead-zone parameter a^* .

The minimization problem in Step 3 is generally non-differentiable, and therefore, gradient-based optimization methods such as Newton's method and quasi-Newton methods are not applicable. Moreover, we need to use a global search method for solving the optimization problem since it is generally a nonconvex problem. For this type of problem, an optimization technique of stochastic multi-point search is effective. In this paper, we use the CMA-ES algorithm⁽⁴⁾ to solve the problem. While the CMA-ES algorithm does not require any gradient information, it is similar to quasi-Newton methods in the sense that a covariance matrix closely related to the inverse Hessian is estimated from the information of the objective function values. In the next section, we present the CMA-ES algorithm for minimizing the function (6).

4. CMA-ES

The CMA-ES algorithm is based on an evolution strategy where search points are generated according to a multivariate normal distribution, and the corresponding covariance matrix is updated through the evaluation of the search points. Like particle swarm optimization (PSO), the CMA-ES algorithm does not require that the objective function be differentiable. The PSO algorithm is relatively well known in the control engineering community; see, for instance,⁽²⁾ and⁽⁸⁾. On the other hand, there are only a few applications of the CMA-ES to control problems. In⁽⁹⁾, it is demonstrated that the CMA-ES algorithm outperforms the PSO one through some control problems.

In this kind of heuristic optimization techniques, it is an important aspect to take into account a tradeoff between exploration and exploitation in the process of finding a solution. While high exploration gives high possibility of finding the global solution, high exploitation leads to high convergence speed. To construct a good algorithm, we usually need to adjust some tuning parameters in such an algorithm, and therefore, a small number of such tuning parameters is desirable. In this sense, the CMA-ES is practical and tractable because the CMA-ES is well-tuned and there are few tuning parameters left to the user.

The algorithm mainly consists of the generation of search points, the update of the mean of the search points, the update of the covariance matrix of the search distribution, and the update of the step size parameter.

The dimension of the search space is denoted by n . The outline of the CMA-ES algorithm is concisely but

completely shown as follows.

CMA-ES Algorithm

Step 0. (Setting of parameters) Set the number of search points λ , the number of selected search points μ , the weights $w_i, i = 1, \dots, \mu$, and the other parameters $\mu_{\text{eff}}, \mu_{\text{cov}}, d_\sigma, c_\sigma, c_c, c_{\text{cov}}$. (We will explain how to determine these parameters later.)

Step 1. (Initialization) Initialize the generation number $g = 0$, the covariance matrix $\mathbf{C}^{(0)} = \mathbf{I}$, the step size parameter $\sigma^{(0)} = 0.5$, and the evolution path parameters $\mathbf{p}_\sigma = \mathbf{0}, \mathbf{p}_c = \mathbf{0}$. Set the initial mean $\mathbf{m}^{(0)} \in \mathbb{D}$ where $\mathbb{D} \subseteq \mathbb{R}^n$ is an initial region.

Step 2. (Generation of search points) Search points are generated according to the normal distribution as follows:

$$\mathbf{x}_j^{(g+1)} \sim \mathcal{N}(\mathbf{m}^{(g)}, (\sigma^{(g)})^2 \mathbf{C}^{(g)}) \text{ for } j = 1, \dots, \lambda.$$

Step 3. (Update of the mean) Evaluate the values of the performance index $J(\mathbf{x}_j^{(g+1)})$ for the search points, and select μ best points $\mathbf{x}_{i:\lambda}^{(g+1)}, i = 1, \dots, \mu$. Update the mean of the search distribution as follows:

$$\mathbf{m}^{(g+1)} = \sum_{i=1}^{\mu} w_i \mathbf{x}_{i:\lambda}^{(g+1)} \dots \dots \dots (7)$$

Step 4. (Update of the covariance matrix) Update the covariance matrix as follows:

$$\begin{aligned} \mathbf{C}^{(g+1)} &= (1 - c_{\text{cov}}) \mathbf{C}^{(g)} + c_{\text{cov}} \left(1 - \frac{1}{\mu_{\text{cov}}} \right) \\ &\cdot \sum_{i=1}^{\mu} w_i \left(\frac{\mathbf{x}_{i:\lambda}^{(g+1)} - \mathbf{m}^{(g)}}{\sigma^{(g)}} \right) \left(\frac{\mathbf{x}_{i:\lambda}^{(g+1)} - \mathbf{m}^{(g)}}{\sigma^{(g)}} \right)^T \\ &+ \frac{c_{\text{cov}}}{\mu_{\text{cov}}} \left(\mathbf{p}_c^{(g+1)} \mathbf{p}_c^{(g+1)T} + \delta(H_\sigma) \mathbf{C}^{(g)} \right), \dots \dots \dots (8) \end{aligned}$$

where

$$\begin{aligned} \mathbf{p}_c^{(g+1)} &= (1 - c_c) \mathbf{p}_c^{(g)} \\ &+ H_\sigma \sqrt{c_c(2 - c_c)\mu_{\text{cov}}} \frac{\mathbf{m}^{(g+1)} - \mathbf{m}^{(g)}}{\sigma^{(g)}}, \dots \dots \dots (9) \end{aligned}$$

$$\delta(H_\sigma) = (1 - H_\sigma)c_c(2 - c_c), \dots \dots \dots (10)$$

$$H_\sigma = \begin{cases} 1 & \text{if } \frac{\|\mathbf{p}_\sigma\|}{\sqrt{1 - (1 - c_\sigma)^{2(g+1)}}} < (1.4 + \frac{2}{n+1}) \mathbb{E}\|\mathcal{N}(\mathbf{0}, \mathbf{I})\| \\ 0 & \text{otherwise,} \end{cases} \dots \dots \dots (11)$$

$$\mathbb{E}\|\mathcal{N}(\mathbf{0}, \mathbf{I})\| = \sqrt{n} \left(1 - \frac{1}{4n} + \frac{1}{21n^2} \right) \dots \dots \dots (12)$$

Step 5. (Update of the step size parameter) Update the step size parameter as follows:

$$\sigma^{(g+1)} = \sigma^{(g)} \exp \left(\frac{c_\sigma}{d_\sigma} \left(\frac{\|\mathbf{p}_\sigma^{(g+1)}\|}{\mathbb{E}\|\mathcal{N}(\mathbf{0}, \mathbf{I})\|} - 1 \right) \right), \dots \dots \dots (13)$$

where

$$\mathbf{p}_\sigma^{(g+1)} = (1 - c_\sigma) \mathbf{p}_\sigma^{(g)} + \sqrt{c_\sigma(2 - c_\sigma)\mu_{\text{eff}}} \left(\mathbf{C}^{(g)} \right)^{-\frac{1}{2}} \cdot \frac{\mathbf{m}^{(g+1)} - \mathbf{m}^{(g)}}{\sigma^{(g)}} \quad (14)$$

Step 6. (Stopping criterion) If a stopping criterion such as the maximum number of generation is satisfied, stop. Otherwise, set $g = g + 1$ and go to Step 2.

Although the above algorithm includes some parameters to be set in Step 0, these parameters including even the number of search points λ are automatically determined in the CMA-ES algorithm developed from theoretical and empirical viewpoints; see ⁽⁴⁾ for detail. Therefore, note that the CMA-ES algorithm does not need so much effort for parameter tuning. Basically, we adopt such parameter settings in the CMA-ES algorithm. The parameters are determined as follows.

$$\lambda = 4 + \lfloor 3 \ln n \rfloor, \quad (15)$$

$$\mu = \lceil (\lambda - 1)/2 \rceil, \quad (16)$$

$$w_i = \frac{w'_i}{\sum_{i=1}^{\mu} w'_i}, \quad (17)$$

$$w'_i = \ln((\lambda - 1)/2 + 1) - \ln i, \quad (18)$$

$$\mu_{\text{eff}} = \frac{1}{\sum_{i=1}^{\mu} w_i^2}, \quad (19)$$

$$\mu_{\text{cov}} = \mu_{\text{eff}}, \quad (20)$$

$$d_\sigma = 1 + 2 \max \left(0, \sqrt{\frac{\mu_{\text{eff}} - 1}{n + 1}} - 1 \right) + c_\sigma, \quad (21)$$

$$c_\sigma = \frac{\mu_{\text{eff}} + 2}{n + \mu_{\text{eff}} + 3}, \quad (22)$$

$$c_c = \frac{4}{n + 4}, \quad (23)$$

$$c_{\text{cov}} = \frac{1}{\mu_{\text{cov}}} \frac{2}{(n + \sqrt{2})^2} + \left(1 - \frac{1}{\mu_{\text{cov}}} \right) \min \left(1, \frac{2\mu_{\text{cov}} - 1}{(n + 2)^2 + \mu_{\text{cov}}} \right). \quad (24)$$

Here, we give some comments on the parameter settings. In Step 4 of the algorithm, the covariance matrix is updated mainly based on the previous covariance matrix, the current covariance matrix of the μ best points, and transition information $\mathbf{p}_c^{(g+1)}$ on the distribution mean $\mathbf{m}^{(g)}$. These three kinds of information are weighted by the parameters c_{cov} and μ_{cov} . The parameter d_σ is a damping factor that scales the change magnitude of $\ln \sigma^{(g)}$. The parameters c_σ and c_c are weighting factors to be used for calculating $\mathbf{p}_c^{(g+1)}$ and $\sigma^{(g+1)}$, respectively. Although μ_{cov} can be selected independently from μ_{eff} , $\mu_{\text{cov}} = \mu_{\text{eff}}$ is often appropriate. For more detailed explanation on the parameter settings; see ⁽⁴⁾.

5. USM Control System

We apply the proposed method to a USM control system (Canon UA60) and verify its effectiveness. In this

section, we first describe the features of USMs and some problems of the control methods of the USMs. Then, we present the configuration of the experimental USM control system and its dead-zone property.

5.1 Features and Control Methods of USMs

As is well known, USMs have excellent features such as small size, light weight, no driving sound, high torque at even low speed, and high holding torque. In addition, USMs produce no electromagnetic noise in principle and are not influenced by the electromagnetic field. Because of this electromagnetic compatibility, USMs are promising devices in application fields such as medical and welfare fields as described in ⁽³⁾.

However, it is not easy to control USMs on the basis of its physical analysis because the USMs are driven by the friction force in principle. Although some plant models for USMs have been proposed, they are quite complicated and impractical for control. In contrast, PID control can be constructed with no plant model and has therefore been used widely as a design scheme for USMs.

Moreover, the characteristics of the USMs are sensitive to aging, ambient temperature, load, and so on, which leads to deterioration of control performance under a fixed-gain PID control. To overcome this problem and constantly achieve good performance, some control methods for tuning PID gains, such as genetic algorithms (GAs) in ⁽¹⁰⁾ and fuzzy inference in ⁽⁵⁾, have been proposed. However the GAs are relatively complicated to implement for online PID controller tuning. One of the reasons is that we have to determine how to encode the real numbers of the PID gains in advance. Further, in the fuzzy inference of ⁽⁵⁾, we need some information on the dead-zone from pre-experiments, in order to obtain appropriate membership functions. One of the advantages of the proposed method is that it is simpler than the abovementioned methods.

5.2 System Configuration The configuration of the experimental USM control system considered in this study is shown in Fig. 3. The USM and encoder are connected to the same shaft. The position information of the USM is obtained by the encoder and is transmitted to a counter board installed in a personal computer (PC). The information on the control input calculated from the output and reference signal in the PC is transmitted to the drive circuit through an I/O board. Fig. 4 shows the drive circuit. The phase difference control circuit consists of digital elements using shift registers, and the variable voltage and booster circuit consists of a

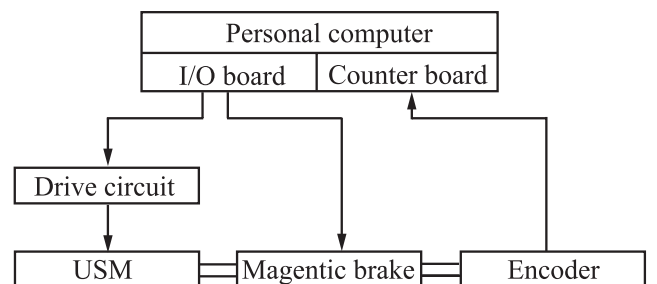


Fig. 3. USM control system.

digital potentiometer, power operational amplifier, and step-up transformer. The starting frequency is fixed at 36.0 kHz, which is 1.0 kHz higher than the resonant frequency of the USM, so as to start the USM securely. The USM is controlled by a phase difference scheme that shows relatively weak hysteresis and an almost linear input/output property. The phase difference can be adjusted between -90 and 90° in steps of 1.406° . A load of 0.1 N·m can be generated by an electromagnetic brake with a supply voltage of 12 V. Table 1 shows the specifications of the USM, encoder, and load. Table 2 shows the specifications of the drive circuit.

5.3 Dead-Zone of USM Fig. 5 shows both ve-

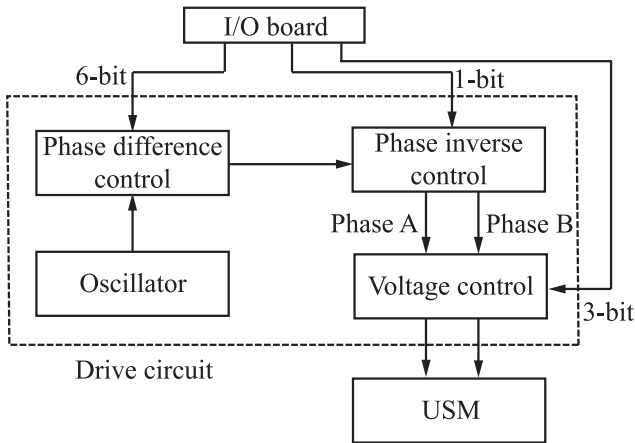


Fig. 4. Drive circuit.

Table 1. Specifications of USM, encoder, and load.

USM	Rated rotational speed: 100 rpm
	Rated torque: 0.392 N·m
	Holding torque: 0.392 N·m
Encoder	Resolution: 0.0036°
Load (Magnetic brake)	0.1 N·m

Table 2. Specifications of drive circuit.

Driving frequency	36.0 kHz
Driving voltage	150 V
Phase difference	$-90 \sim 90^\circ$
Resolution of phase difference	1.406°
Sampling time	1.0 ms

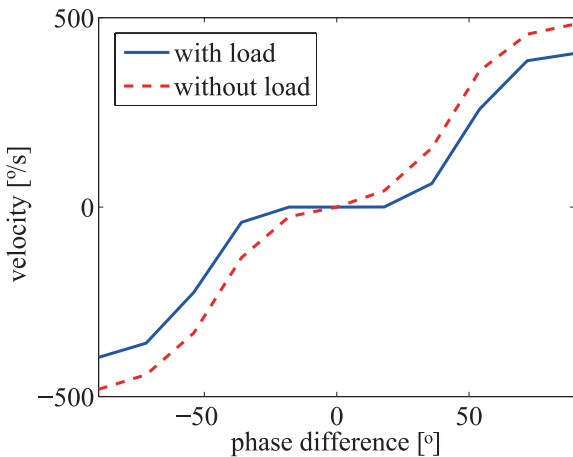


Fig. 5. Velocity vs. phase difference.

locities of the USM with and without the load. We can observe from this figure that the dead-zone for the input varies depending on the existence of the load. In fact, the dead-zone is sensitive to the strength of the load, and therefore, we have to compensate for the dead-zone easily in order to achieve good control performance, particularly when it is either impossible or difficult to estimate the strength of the load.

6. Experimental Results

In this section, we present some experimental results by the FRIT methods. For comparison, we carried out experiments both with and without dead-zone compensation. In the conventional FRIT method, *i.e.*, the FRIT method without dead-zone compensation, the tuning parameter $\mathbf{x} = (\theta^T, a)^T = (K_P, K_I, K_D, a)^T$ is replaced with $\mathbf{x} = \theta = (K_P, K_I, K_D)^T$. As described in Subsection 5.2, the control input is the phase difference between the two phases of sinusoidal signals applied to the USM. Therefore, the unit of the control input is degree $[\circ]$. The control output is the position, *i.e.*, the rotation angle of the USM, and its unit is also degree $[\circ]$. The reference model is given by $M(z) = (0.001752z + 0.01534)/(z^2 - 1.637z + 0.6703)$ which is obtained by discretizing the continuous-time system $M(s) = 1/(0.005s + 1)^2$ with sampling time 0.001 s.

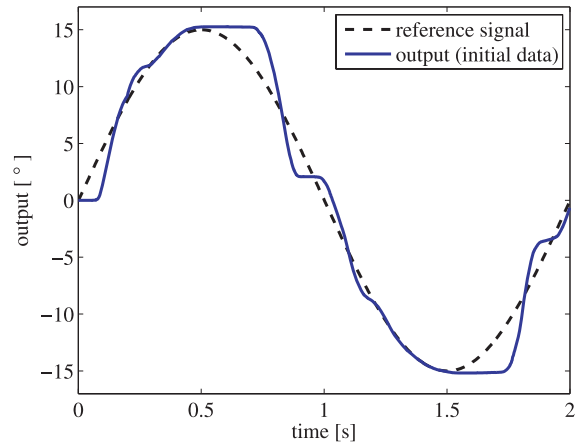


Fig. 6. Initial control output.

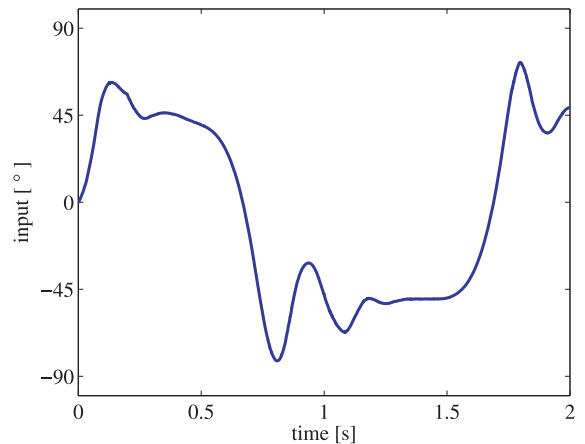


Fig. 7. Initial control input.

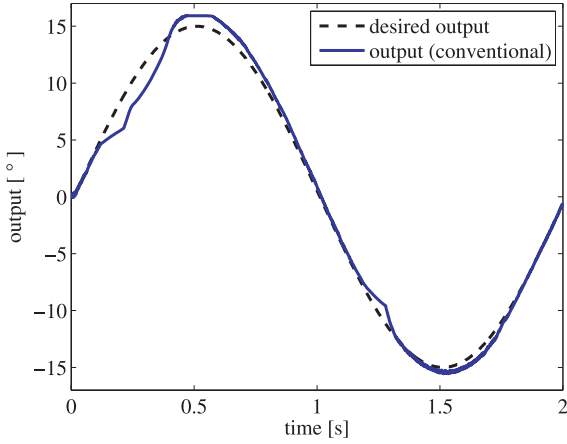


Fig. 8. Control output by the conventional FRIT method with the large search region.

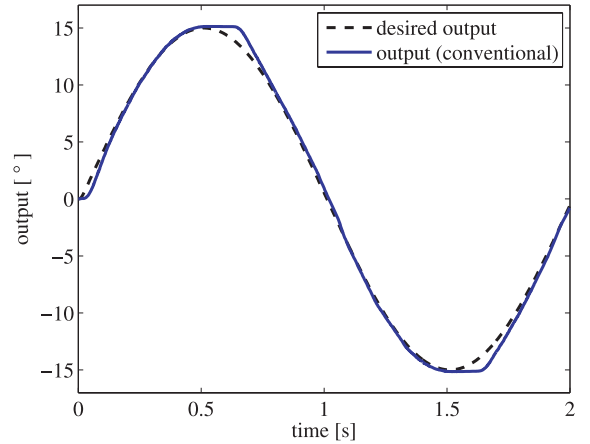


Fig. 10. Control output by the conventional FRIT method with the relatively small search region.

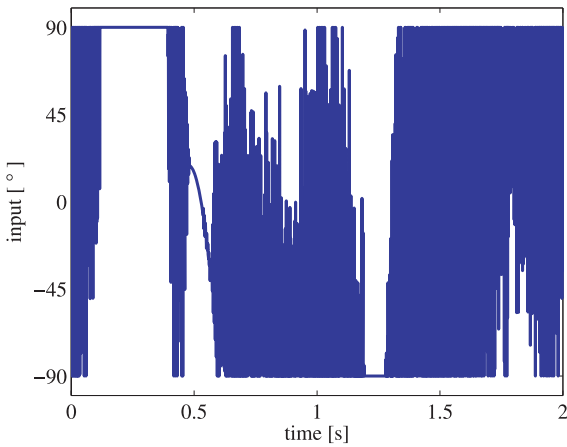


Fig. 9. Control input by the conventional FRIT method with the large search region.

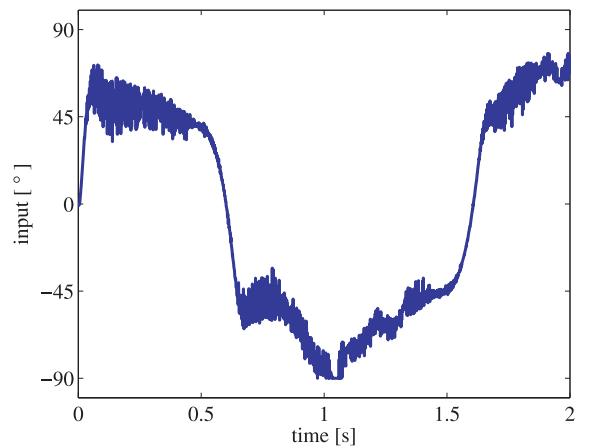


Fig. 11. Control input by the conventional FRIT method with the relatively small search region.

6.1 Example 1 The reference signal is a sinusoidal signal with a magnitude of 15° and a period of 2 s. We set $N = 2000$ to evaluate control performance within 2 s and impose the load. By giving the reference signal $r(k)$ to the closed-loop system with the initial PID gains $\theta_0 = (4, 0.2, 4)^T$, we obtained input/output data without dead-zone compensation as shown in Figs. 6 and 7.

Case 1 At first, we show the experimental results by the conventional FRIT method. We set a sufficiently large search region as $\{\mathbf{x} | \mathbf{x}_{lb} \leq \mathbf{x} \leq \mathbf{x}_{ub}\}$ with $\mathbf{x}_{lb} = (0, 0, 0)^T$ and $\mathbf{x}_{ub} = (400, 4, 2000)^T$, so that the obtained PID gains $\theta^* = (98.8747, 0.1788, 1174.7124)^T$ are not on the boundaries of the search space. In the CME-ES algorithm, the maximum generation number was set to 500. The algorithm programmed in MATLAB was run on a computer with 2.4GHz Core Duo CPU and 1024MB RAM. The computation time was 94.3480 s. The control output and input by the experiment with the obtained PID gains are shown in Figs. 8 and 9, respectively. We see from the figures that the control input is frequently saturated, and the control output does not track the desired output precisely.

Case 2 In order to avoid such saturation in Case 1,

we set the relatively small search region by replacing the upper bound with $\mathbf{x}_{ub} = (40, 4, 200)^T$. Under this setting, we obtained the PID gains $\theta^* = (40, 0.4172, 200)^T$ by the conventional FRIT method. We show the control output and input obtained by implementing the PID gains in Figs. 10 and 11, respectively. In this case, we see that the control output does not track the desired output during low-speed rotation due to dead-zone, while the control input is not saturated almost all the time.

As shown above, it is often difficult to give good control performance by the conventional FRIT method due to the dead-zone and input saturation. Now we show the experimental results by the proposed FRIT method. We set the search region with the lower/upper bounds $\mathbf{x}_{lb} = (0, 0, 0, 0)^T$, $\mathbf{x}_{ub} = (40, 4, 200, 90)^T$ and obtained the PID gains $\theta^* = (32.9815, 0.0256, 150.6710)^T$ and the dead-zone parameter $a^* = 36.5007$. Note that the obtained solutions are not on the boundaries of the search region. The computation time was 104.7950 s. We show the control output and input by the proposed FRIT method in Figs. 12 and 13, respectively. We see from the figures that the control output tracks the desired output precisely while the control input is not saturated.

6.2 Example 2 To show the effectiveness of the

Table 3. Control performance.

	Case 1	Case 2	Case 3	Case 4	Case 5	Case 6	Case 7
Initial experiment	5197.97	5197.97	1641.76	3619.10	2510.16	4799.26	7360.49
Conventional FRIT method	1717.58	543.19	124.87	222.83	299.35	751.91	402.68
Proposed FRIT method	—	86.68	45.99	41.97	103.63	118.28	65.03

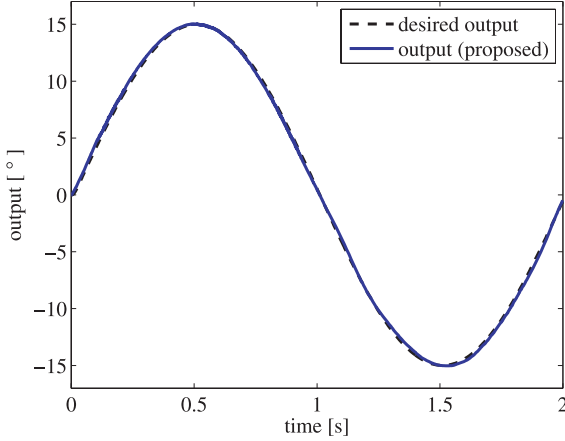


Fig. 12. Control output by the proposed FRIT method.

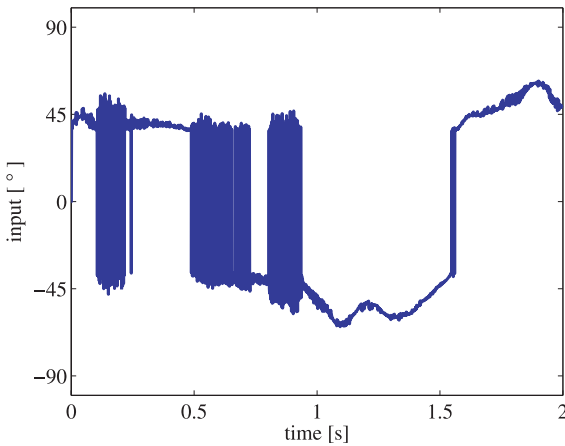


Fig. 13. Control input by the proposed FRIT method.

proposed FRIT method for various cases, we carried out experiments by changing some conditions of Case 2 in Example 1. The control performance is evaluated by the performance index (6) where the initial output $y_0(k)$ is replaced with the resulting output $y(k)$. For each case, we describe the condition changed from Case 2 as follows.

Case 3 The period of the sinusoidal reference signal is changed to 4 s while the evaluated duration is still 2 s as in Case 2.

Case 4 The magnitude of the sinusoidal reference signal is changed to 10° .

Case 5 The load is not imposed.

Case 6 The time constant of the reference model is changed from 0.005 to 0.001.

Case 7 The initial PID gains are changed to $\theta_0 = (10, 0.1, 10)^T$.

In Table 3, we show the performance index values for

Cases 1–7 where $J_0 = \sum_{k=1}^N (y_0(k) - r(k))^2$ is used for the initial experiments. We see from the table that the proposed FRIT method gives better performance than the conventional one for all the cases.

7. Conclusion

In this paper, we have proposed an FRIT method combined with dead-zone compensation and verified its effectiveness by experiments. Although we have dealt with a dead-zone property, a similar approach may be effective for other nonlinearities, which is one of the future research directions.

References

- (1) S. Souma, O. Kaneko, and T. Fujii: “A New Method of Controller Parameter Tuning Based on Input-Output Data, – Fictitious Reference Iterative Tuning”, Proc. IFAC Workshop on Adaptation and Learning in Control and Signal Processing (ALCOSP 04), pp.789–794 (2004)
- (2) K. Tasaka, M. Kano, M. Ogawa, S. Masuda, and T. Yamamoto: “Direct PID Tuning from Closed-Loop Data and Its Application to Unstable Processes”, *Trans. Institute of Systems, Control and Information Engineers*, Vol.22, No.4, pp.137–144 (2009) (in Japanese)
- (3) T. Sashida and T. Kenjo: *An Introduction to Ultrasonic Motors*, Oxford University Press (1994)
- (4) N. Hansen: “The CMA Evolution Strategy: A Tutorial”, available at <http://www.lri.fr/~hansen/cmatutorial.pdf> (2010)
- (5) T. Senjyu, T. Kashiwagi, T. Yoshida, and K. Uezato: “Position Control of Ultrasonic Motors Using Dead-Zone Compensation with Fuzzy Inference”, *Journal of Japan Society for Fuzzy Theory and Systems*, Vol.14, No.2, pp.190–197 (2002) (in Japanese)
- (6) Y. Wakasa, K. Tanaka, T. Akashi, and Y. Nishimura: “PSO-based Simultaneous Tuning Method for PID Controllers and Dead-Zone Compensators and Its Application to Ultrasonic Motors”, *International Journal of Innovative Computing, Information and Control*, Vol.6, No.10, pp. 4593–4604 (2010)
- (7) M. Campi and S. Savarese: “Direct nonlinear control design: the virtual reference feedback tuning (VRFT) approach”, *IEEE Trans. Automatic Control*, Vol.51, No.1, pp.14–27 (2006)
- (8) I. Maruta, T.-H. Kim, and T. Sugie: “Fixed-Structure H_∞ Controller Synthesis: A Meta-Heuristic Approach using Simple Constrained Particle Swarm Optimization”, *Automatica*, Vol.45, pp.553–559 (2009)
- (9) Y. Wakasa, S. Kanagawa, K. Tanaka, and Y. Nishimura: “PID Controller Tuning Based on the Covariance Matrix Adaptation Evolution Strategy”, *IEEJ Trans. Electronics, Information and Systems*, Vol.130, No.5, pp.737–742 (2010)
- (10) P. J. Fleming and R. C. Purshouse: “Evolutionary Algorithms in Control Systems Engineering: A Survey”, *Automatica*, Vol.10, pp.1223–1241 (2002)

Yuji Wakasa

(Member) received his B.S. and M.S. degrees in engineering and the degree of doctor of informatics from Kyoto University in 1992, 1994, and 2000. He was a research associate at Okayama University from 1994 to 1998 and at Kyoto University from 1998 to 2002. Since then, he has been with the Graduate School of Science and Engineering at Yamaguchi University. He is currently an associate professor. His current research interests include control system design via mathematical programming and its applications. He is a member of SICE, ISCIE, ORSJ, and IEEE.

Kanya Tanaka

(Senior Member) graduated from the Science and Engineering program at the National Defense Academy in 1982. From 1990 through 1992, he was an associate professor at the Department of Electronic Engineering, the Hiroshima Denki Institute of Technology. From 1992 through 1998, he was an associate professor at the Department of Mechanical Engineering, Ehime University. He is currently a professor of the Graduate School of Science and Engineering at Yamaguchi University. His main interests are in the theory of intelligent control and related applications. He is a member of SICE, ISCIE, JSME, JSAEM, and IEEE.

Shinji Kanagawa

(Non-member) graduated from the Department of Electrical and Electronic Engineering, Yamaguchi University in 2009. He is currently a student at the Graduate School of Science and Engineering, Yamaguchi University. His current research interests include control system design via evolutionary computation.

Yuki Nishimura

(Member) received his B.S., M.S., and Ph.D. degrees from Hokkaido University, Japan, in 2002, 2005, and 2009, respectively. In 2009, he joined the faculty of Yamaguchi University, where he is currently an assistant professor of the Graduate School of Science and Engineering. His research interests include nonlinear control theory. He is a member of SICE, ISCIE, and IEEE.